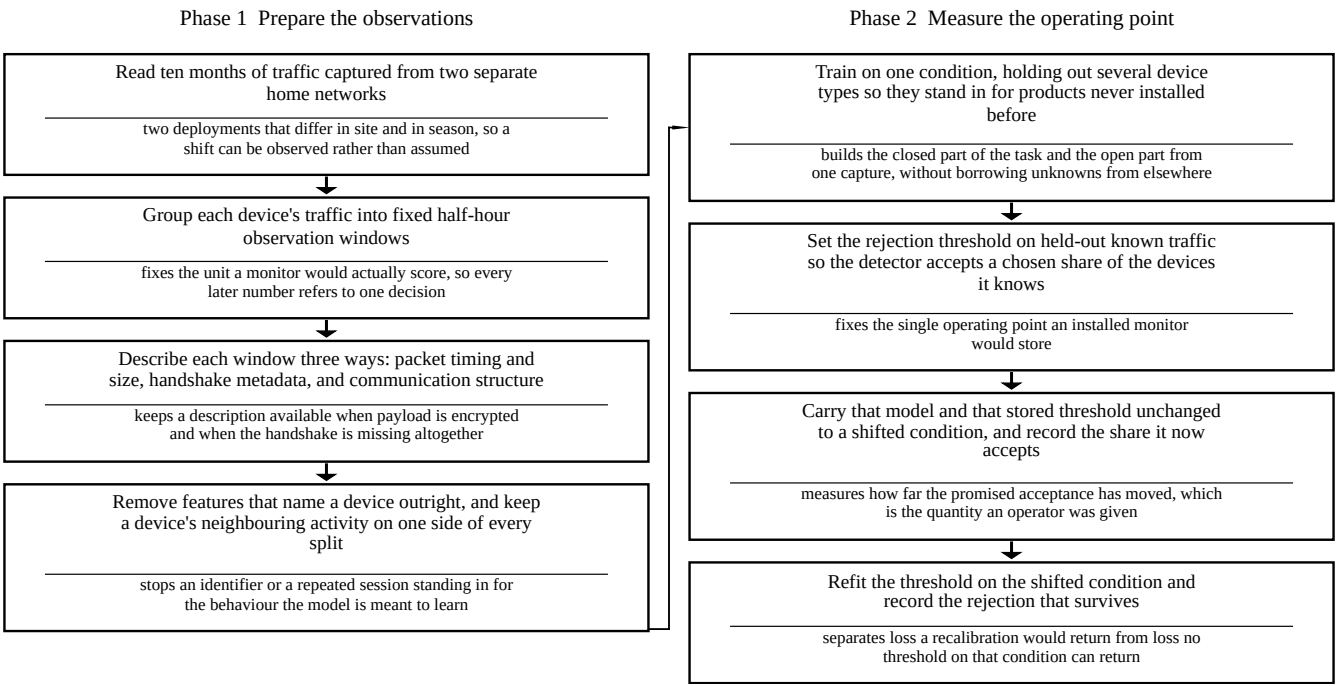
Phase 1 Prepare the observations
Read ten months of traffic captured from two separate home networks
two deployments that differ in site and in season, so a shift can be observed rather than assumed
Group each device's traffic into fixed half-hour observation windows
fixes the unit a monitor would actually score, so every later number refers to one decision
Describe each window three ways: packet timing and size, handshake metadata, and communication structure
keeps a description available when payload is encrypted and when the handshake is missing altogether
Remove features that name a device outright, and keep a device's neighbouring activity on one side of every split
stops an identifier or a repeated session standing in for the behaviour the model is meant to learn
Phase 2 Measure the operating point
Train on one condition, holding out several device types so they stand in for products never installed before
builds the closed part of the task and the open part from one capture, without borrowing unknowns from elsewhere
Set the rejection threshold on held-out known traffic so the detector accepts a chosen share of the devices it knows
fixes the single operating point an installed monitor would store
Carry that model and that stored threshold unchanged to a shifted condition, and record the share it now accepts
measures how far the promised acceptance has moved, which is the quantity an operator was given
Refit the threshold on the shifted condition and record the rejection that survives
separates loss a recalibration would return from loss no threshold on that condition can return